# Benedict Chacko Mathew

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## Summary

Final-year B.Tech Computer Science (Class of 2027, CGPA 8.5/10.0) with strong foundations in Java, OOP, DSA, SQL, REST APIs and Spring Boot, plus hands-on full-stack and AI/ML experience. Open-source contributor to Google Bazel (merged Java security fix, OSVRP-acknowledged) and Jupyter Notebook (merged routing fix, v7.7.0). Internships at Infosys Springboard (ML + full-stack) and Edunet Foundation (Azure + TensorFlow CNN, 92% accuracy); disclosed 15+ vulnerabilities across 20+ bug-bounty programs. Cyber Security Lead mentoring 100+ students. Seeking entry-level Software Engineer roles.

## Education

**Viswajyothi College of Engineering & Technology** **Expected: May 2027**
*B.Tech in Computer Science & Engineering* | ***CGPA: 8.5/10.0*** | *No Active Backlogs* *Vazhakulam, Kerala*

- **Coursework:** DSA, OOP, Java Programming, DBMS, OS, Computer Networks, Software Engineering (SDLC, Agile), ML, AI
- **Class XII (St. George HSS, Muthalakodam):** 95.7% – 2023 | **Class X:** 98.0% – 2021

## Technical Skills

**Languages:** Java, Python, JavaScript/TypeScript (ES6+), SQL
**Core Java:** OOP, Collections, Exception Handling, JDBC, Multithreading & Concurrency, Dependency Injection
**DSA:** Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Hashing, Sorting, Searching
**Backend:** Spring Boot, Spring MVC, REST APIs, Hibernate/JPA, Flask, Microservices, Kafka, gRPC, Redis
**AI/ML:** scikit-learn, TensorFlow, YOLOv8, OpenCV, MediaPipe, NumPy, EDA, Feature Engineering, Cross-Validation
**Databases:** MySQL, PostgreSQL, SQL Server, MongoDB, Normalization, Query Optimization
**Frontend:** React.js, HTML5, CSS3, Tailwind CSS, Vite, TypeScript
**Tools & Cloud:** Git, GitHub, Maven, Postman, Docker, Playwright, Azure (ML Studio, Data Factory), Burp Suite, OWASP ZAP
**Practices:** SDLC, Agile/Scrum, Debugging, JUnit, Regression Testing, API Documentation, OWASP Top 10

## Open Source Contribution

**Google Bazel** **2026**
*Open Source Contributor & Security Researcher* *Remote*

- Fixed **CWE-522 credential leak** in Bazel's **ProxyHelper** (Java) with host-port-scoped **ConcurrentHashMap** + requestor fallback for thread-safe auth
- Added **JUnit tests** for isolation and teardown; merged into **bazelbuild/bazel (dc50161)**; acknowledged by Google **OSVRP**

**Jupyter Notebook** **2026**
*Open Source Contributor* *Remote*

- Fixed tree-view misrouting where paths containing a **notebooks** folder were intercepted as file-open routes (404 File Load Error) by anchoring opener route regex in **TypeScript** application extension
- Added **Playwright regression test** for navigate + reload on nested tree paths; merged into **jupyter/notebook (7903)**, milestone **7.7.0**, fixing issue **7216**

## Work Experience

**AI/ML Intern** **Feb 2026 – Apr 2026**
*Infosys Springboard* *Remote*

- Developed **visa-approval classifier** on **25,480-record EasyVisa** dataset (23 features); EDA, preprocessing and feature engineering; model selection with **5-fold CV**, shipped **Random Forest** as best model
- Exposed model via **Flask REST APIs** with **React 18 + Vite** analytics UI; Agile delivery with API documentation

**AI & Azure Cloud Intern** **Jun 2025 – Aug 2025**
*Edunet Foundation* *Remote*

- Built ML lifecycle on **Azure (ML Studio, Data Factory)** – versioning and retraining; **TensorFlow CNN**: **92% accuracy**, **20% faster inference**
- Automated Python ingestion pipeline for **100K+ images**; cloud deployment with reporting

**Freelance Web Developer & Security Researcher** **Jan 2023 – Present**
*Independent* *Remote*

- Delivered responsive web apps with **React, JavaScript and Vue.js** for clients
- Disclosed **15+ vulnerabilities (XSS, CSRF, IDOR, SQLi)** following **OWASP Top 10**; security-tested **20+ programs** with **Burp Suite and OWASP ZAP**

## Project Experience

**SentinelMesh – Infra Monitoring Platform** | **Java, Spring Boot, Kafka, gRPC** **2026**

- Architected **Spring Boot microservices** with **Kafka, Redis, PostgreSQL, gRPC and Docker**; 60-sec correlation windows with SSE live dashboards; validated on **350+ synthetic events at <10s end-to-end latency**
- Added **Isolation Forest + rule engine** for DDoS, auth-abuse and resource-leak detection; applied DI, concurrency and **JUnit** testing

**Silent Invigilator – Exam Malpractice Detection** | **Python, YOLOv8, OpenCV** **2026**

- Built real-time detector flagging **phone use, gaze deviation, multi-person frames and head movement** via **YOLOv8**, face counting, 3D head pose and iris tracking (**OpenCV, MediaPipe, NumPy**)
- Added anomaly scoring with structured **JSON reports** for review

**Visa Prediction Platform** | **Python, Flask, React** **Mar 2026**

- Production-style full-stack app serving sklearn model via **Flask REST APIs**; **React 18 + Vite** UI with real-time predictions, confidence scores and trend analytics; deployed on **Netlify/Vercel** *(capstone from Infosys Springboard internship above)*

## Certifications

- **Applied Generative AI** – Infosys Springboard | 2025 | **OpenAI GPT-3** – Infosys Springboard | 2025
- **Google Cloud Foundations & GenAI** – Google Cloud Skills Boost | 2025 | **Power Automate** – Microsoft Learn | 2025

## Achievements & Leadership

- **Cyber Security Lead, MuLearn** – led peer learning for **100+ students**; conducted **5+ workshops**; talks on security, bug bounty and blockchain
- Mentored **100+ peers** in hackathons and coding contests (DSA, Git, Agile)